Sinh Hoang

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KEY SKILLS

Swift, SwiftUI, Combine, UIKit, RxSwift

WORK EXPERIENCE

Aviator App **January 2025 - Now**
iOS Engineer *Ha Noi, Viet Nam*

- An airline flight operations app (UIKit, Swift) covering flight briefing, NavLog, weather, dispatch, and pilot utilities.
- Maintain and extend protocol-based dependency injection wrappers around system APIs (NotificationCenter, FileManager, UIApplication, BGTaskScheduler) to keep code testable and mockable.
- Work on operator-configurable features driven by XML-based configuration, accessed through a generated dot-notation config accessor.
- Contribute to flight plan workflow services integrating with EFF and flight planning systems, and CoreData-backed persistence.
- Implement and maintain features across Dashboard, Weather, and Flight Briefing modules, following existing architecture and logging conventions.
- Migrate to Swift 6 with strict concurrency enabled, resolving data race warnings and adopting modern concurrency-safe patterns across the codebase.
- Migrate dependency management from Carthage to Swift Package Manager (SPM), streamlining build times and simplifying dependency resolution.
- Write unit tests using mocks and model factories, following existing test conventions to keep coverage high.
- Represent the Vietnam team as a key point of contact to coordinate with the global team, driving alignment and resolving cross-team blockers.
- Technologies: Swift, UIKit, CoreData, XCTest, Carthage, SPM, XML configuration.

Smart Home app **August 2023 - December 2024**
iOS Engineer *Ha Noi, Viet Nam*

- An application that helps users manage smart devices in their homes.
- Define a scalable app architecture to facilitate seamless integration of new device types and features.
- Create a robust core layer for discovering, connecting, and interacting with various smart devices.
- Implement automation features, enabling users to create schedules and triggers for device actions.
- Add real-time push notifications for events.
- Implement multi-user support with role-based permissions for shared device access.
- Develop a seamless firmware update system to ensure that connected devices remain up-to-date.
- Write automation tests to test the app's performance.
- Technologies: SwiftUI, Combine, Realms, AWS, Core Bluetooth, XCTest, Appium.

MoneyLover App **January 2020 - July 2023**
iOS Engineer *Ha Noi, Viet Nam*

- An application that helps users manage their personal finances.

- Responsible for making Onboarding, Tutorial, Budget Management, Home, and In-app purchase features.
- Responsible for developing and releasing the first version of WatchOS, and implementing Home Screen and Lock Screen Widgets.
- Add a new feature to the current CI/CD pipeline to enable developers to hide production API endpoints.
- Maintain legacy Objective-C code while developing new features in Swift.
- Research SwiftUI and then apply it to new features.
- Technologies: Swift, SwiftUI, RxSwift, SQLite, Firebase, Charts, Snapkit, Tracking Service.

OTHER PROJECTS

E2EE Messaging App **May 2025 - Now**
https://github.com/sinhlhhn/E2EE-Messaging-App *Ha Noi, Viet Nam*

- A secure, modular, and real-time messaging system with end-to-end encryption and live communication features.
- Create a scalable architecture with Composition Root, Decorator pattern, Dependency Injection.
- Implement robust authentication layer with token management and auto-refresh.
- Implement secure, real-time chat functionality with end-to-end encryption, key agreement mechanisms, SSL pinning (public key), and secure key storage.
- Implement secure media sharing (images, videos, and various file types) with encryption and access control.
- Build a Node.js/Express server to support real-time communication, store media files and persistent chat history using cursor-based pagination.
- Technologies: SwiftUI, Combine, SocketIO, NodeJs, SQLite.

Essential App **July 2023 - November 2023**
https://github.com/sinhlhhn/EssentialApp *Ha Noi, Viet Nam*

- An application that helps users see the Feed and the comments.
- Draw dependencies diagrams to keep track of dependencies.
- Follow TDD approach.
- Apply different testing strategies: Unit test, Integration test, UI test, Snapshot test.
- Separate modules to achieve the modular design.
- Apply Design pattern to practice (Composite, Decorator, Adapter, Strategy pattern).
- Set up Github Action CI/CD.
- Technologies: Swift, XCTest

AWARDS

Best Performance Award *FPT*
For Outstanding Performance *December 2024*

Raising Star Award *MoneyLover*
For Outstanding Performance and Dedicated Service *December 2022*

EDUCATION

EnglishScore *September 2024*
B2-Upper Intermediate

iOS Lead Essentials *November 2023*
Essential Developer Academy

Bachelor's degree in Information Technology *Hanoi University of Industry*
Bachelor's degree program *Sep 2017 - June 2021*
Very Good. GPA: 3,4/4

FAVORITE BOOK

Dependency Injection *Steven van Deursen and Mark Seemann*
Principles, Practices, Patterns

A Philosophy of Software Design *John Ousterhout*